SPACE PROGRAMME: ISRO, ORGANISATION AND LAUNCH VEHICLES

INSTITUTIONAL COMMAND MAP → CENTRE-TO-LAUNCH CHAIN → POLICY AND LAW STACK → LAUNCHER-PAYLOAD SEPARATION → ORBIT VOCABULARY MATRIX → GTO-TO-GEO RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure: the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00

INSTITUTIONAL COMMAND MAP

INSTITUTIONAL COMMAND MAP · DEPARTMENT OF SPACE · POLICY UMBRELLA AND SOVEREIGN DIRECTION · PUBLIC R&D, LAUNCH VEHICLES, SPACECRAFT AND MISSIONS · PROMOTE, AUTHORISE AND SUPERVISE NON-GOVERNMENT ACTIVITY · COMMERCIAL CONTRACTS, SERVICES AND TECHNOLOGY TRANSFER · INSTITUTION NAMES ARE NOT INTERCHANGEABLE

DEPARTMENT OF SPACE → policy umbrella and sovereign direction

ISRO → public R&D, launch vehicles, spacecraft and missions → IN-SPACe → promote, authorise and supervise non-government activity → NSIL → commercial contracts, services and technology transfer

CONCEPT / ARCHITECTURE
- DEPARTMENT OF SPACE → policy umbrella and sovereign direction
- ISRO → public R&D, launch vehicles, spacecraft and missions

MECHANISM / APPLICATION
- IN-SPACe → promote, authorise and supervise non-government activity
- NSIL → commercial contracts, services and technology transfer

READINESS / STATUS LIMIT
- RULE → institution names are not interchangeable
- DEPARTMENT OF SPACE → policy umbrella and sovereign direction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NSIL → commercial contracts, services and technology transfer.

01

STAGE 01

CENTRE-TO-LAUNCH CHAIN

CENTRE-TO-LAUNCH CHAIN · LAUNCH-VEHICLE DESIGN AND DEVELOPMENT · LIQUID PROPULSION AND PROPULSION TESTING ROLES · INTEGRATION, RANGE AND LAUNCH OPERATIONS · MISSION CONTROL · FLIGHT MONITORING AND DECISION CHAIN · A CENTRE FUNCTION IS NOT THE WHOLE ISRO MANDATE

VSSC → launch-vehicle design and development → LPSC / IPRC → liquid propulsion and propulsion testing roles → SDSC-SHAR → integration, range and launch operations → MISSION CONTROL → flight monitoring and decision chain → RULE → a centre function is not the whole ISRO mandate

STARTING CONDITION / INPUT
- VSSC → launch-vehicle design and development
- LPSC / IPRC → liquid propulsion and propulsion testing roles

SCIENTIFIC OR ENGINEERING MECHANISM
- SDSC-SHAR → integration, range and launch operations
- MISSION CONTROL → flight monitoring and decision chain

OUTPUT / FAILURE BOUNDARY
- RULE → a centre function is not the whole ISRO mandate
- VSSC → launch-vehicle design and development

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a centre function is not the whole ISRO mandate.

02

STAGE 02

POLICY AND LAW STACK

POLICY AND LAW STACK · OUTER-SPACE TREATIES · INTERNATIONAL STATE OBLIGATIONS · INDIAN SPACE POLICY 2023 · EXECUTIVE ROLE ALLOCATION · IN-SPACE NORMS · AUTHORISATION AND SUPERVISION INTERFACE · COMMERCIAL CONTRACT

OUTER-SPACE TREATIES → international State obligations

INDIAN SPACE POLICY 2023 → executive role allocation → IN-SPACe NORMS → authorisation and supervision interface → COMMERCIAL CONTRACT → service and risk allocation

CONCEPT / ARCHITECTURE
- OUTER-SPACE TREATIES → international State obligations
- INDIAN SPACE POLICY 2023 → executive role allocation

MECHANISM / APPLICATION
- IN-SPACe NORMS → authorisation and supervision interface
- COMMERCIAL CONTRACT → service and risk allocation

READINESS / STATUS LIMIT
- GAP → policy is not a comprehensive domestic space statute
- OUTER-SPACE TREATIES → international State obligations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GAP → policy is not a comprehensive domestic space statute.

03

STAGE 03

LAUNCHER-PAYLOAD SEPARATION

LAUNCHER-PAYLOAD SEPARATION · LAUNCH VEHICLE · SUPPLIES ASCENT ENERGY AND GUIDANCE · PAYLOAD FAIRING · PROTECTS THE MISSION PAYLOAD · LAUNCHER TASK AND SPACECRAFT TASK DIVERGE · CARRIES PAYLOAD AND MAY RAISE OR MAINTAIN ORBIT · ROCKET SUCCESS AND SPACECRAFT SERVICE ENTRY ARE SEPARATE

LAUNCH VEHICLE → supplies ascent energy and guidance → PAYLOAD FAIRING → protects the mission payload → SEPARATION → launcher task and spacecraft task diverge → SPACECRAFT → carries payload and may raise or maintain orbit → RULE → rocket success and spacecraft service entry are separate

STARTING CONDITION / INPUT
- LAUNCH VEHICLE → supplies ascent energy and guidance
- PAYLOAD FAIRING → protects the mission payload

SCIENTIFIC OR ENGINEERING MECHANISM
- SEPARATION → launcher task and spacecraft task diverge
- SPACECRAFT → carries payload and may raise or maintain orbit

OUTPUT / FAILURE BOUNDARY
- RULE → rocket success and spacecraft service entry are separate
- LAUNCH VEHICLE → supplies ascent energy and guidance

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → rocket success and spacecraft service entry are separate.

04

STAGE 04

ORBIT VOCABULARY MATRIX

ORBIT VOCABULARY MATRIX · ALTITUDE CLASS, OFTEN USED BY CREW AND EARTH OBSERVATION · HIGH-INCLINATION GROUND-TRACK GEOMETRY · REPEATED LOCAL-SOLAR-TIME CROSSING · ORBITAL PERIOD TIED TO EARTH'S ROTATION · CIRCULAR EQUATORIAL GEOSYNCHRONOUS SPECIAL CASE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
LEO → altitude class, often used by crew and Earth observation	SUN-SYNCHRONOUS → repeated local-solar-time crossing	GEO → circular equatorial geosynchronous special case
POLAR → high-inclination ground-track geometry	GSO → orbital period tied to Earth's rotation	LEO → altitude class, often used by crew and Earth observation

SYSTEM / DEVICE / INSTITUTION
- LEO → altitude class, often used by crew and Earth observation
- POLAR → high-inclination ground-track geometry

DECISIVE TECHNICAL DISTINCTION
- SUN-SYNCHRONOUS → repeated local-solar-time crossing
- GSO → orbital period tied to Earth's rotation

STATUS / UNIT / EVIDENCE LIMIT
- GEO → circular equatorial geosynchronous special case
- LEO → altitude class, often used by crew and Earth observation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GSO → orbital period tied to Earth's rotation.

05

STAGE 05

GTO-TO-GEO RAIL

GTO-TO-GEO RAIL · LAUNCHER ASCENT · REACHES TRANSFER CONDITIONS · GTO INJECTION · ELLIPTICAL INTERMEDIATE ORBIT · SPACECRAFT PROPULSION · RAISES PERIGEE AND CIRCULARISES · GEO SLOT

LAUNCHER ASCENT → reaches transfer conditions → GTO INJECTION → elliptical intermediate orbit → SPACECRAFT PROPULSION → raises perigee and circularises → GEO SLOT → circular equatorial final service orbit → TRAP → GTO delivery is not direct GEO placement

STARTING CONDITION / INPUT
- LAUNCHER ASCENT → reaches transfer conditions
- GTO INJECTION → elliptical intermediate orbit

SCIENTIFIC OR ENGINEERING MECHANISM
- SPACECRAFT PROPULSION → raises perigee and circularises
- GEO SLOT → circular equatorial final service orbit

OUTPUT / FAILURE BOUNDARY
- TRAP → GTO delivery is not direct GEO placement
- LAUNCHER ASCENT → reaches transfer conditions

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → GTO delivery is not direct GEO placement.

06

STAGE 06

OPERATIONAL LAUNCHER MATRIX

OPERATIONAL LAUNCHER MATRIX · VERSATILE WORKHORSE AND MIXED MISSION PROFILES · CRYOGENIC UPPER STAGE AND TRANSFER-ORBIT ROLE · HEAVIER OPERATIONAL CLASS AND HUMAN-RATING BASE · SEPARATE SMALL-SATELLITE LAUNCH ARCHITECTURE · ROLE, PROPULSION, ORBIT AND STATUS MUST TRAVEL TOGETHER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PSLV → versatile workhorse and mixed mission profiles	LVM3 → heavier operational class and human-rating base	RULE → role, propulsion, orbit and status must travel together
GSLV → cryogenic upper stage and transfer-orbit role	SSLV → separate small-satellite launch architecture	PSLV → versatile workhorse and mixed mission profiles

SYSTEM / DEVICE / INSTITUTION
- PSLV → versatile workhorse and mixed mission profiles
- GSLV → cryogenic upper stage and transfer-orbit role

DECISIVE TECHNICAL DISTINCTION
- LVM3 → heavier operational class and human-rating base
- SSLV → separate small-satellite launch architecture

STATUS / UNIT / EVIDENCE LIMIT
- RULE → role, propulsion, orbit and status must travel together
- PSLV → versatile workhorse and mixed mission profiles

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → role, propulsion, orbit and status must travel together.

07

STAGE 07

LINEAGE AND STATUS LADDER

LINEAGE AND STATUS LADDER · EXPERIMENTAL ORBITAL-ACCESS LEARNING · INTERMEDIATE TECHNOLOGY-LEARNING STEP · OPERATIONAL FAMILIES IN OWNER · DEVELOPED SMALL-LAUNCH FAMILY WITH DATED STATUS NEEDED · APPROVED FUTURE DEVELOPMENT, NOT AN OPERATIONAL LAUNCHER

SLV-3 → experimental orbital-access learning → ASLV → intermediate technology-learning step → PSLV / GSLV / LVM3 → operational families in owner → SSLV → developed small-launch family with dated status needed → NGLV → approved future development, not an operational launcher

STARTING CONDITION / INPUT
- SLV-3 → experimental orbital-access learning
- ASLV → intermediate technology-learning step

SCIENTIFIC OR ENGINEERING MECHANISM
- PSLV / GSLV / LVM3 → operational families in owner
- SSLV → developed small-launch family with dated status needed

OUTPUT / FAILURE BOUNDARY
- NGLV → approved future development, not an operational launcher
- SLV-3 → experimental orbital-access learning

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NGLV → approved future development, not an operational launcher.

08

STAGE 08

PROPULSION COMPARISON

PROPULSION COMPARISON · STORED PROPELLANT AND LIMITED THROTTLE CONTROL · EARTH-STORABLE LIQUID · AMBIENT-TEMPERATURE LIQUID PAIR · LIQUID OXYGEN PLUS LIQUID HYDROGEN · LIQUID OXYGEN PLUS KEROSENE · PROPELLANT CLASS DOES NOT PROVE FLIGHT STATUS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SOLID → stored propellant and limited throttle control	CRYOGENIC → liquid oxygen plus liquid hydrogen	RULE → propellant class does not prove flight status
EARTH-STORABLE LIQUID → ambient-temperature liquid pair	SEMI-CRYOGENIC → liquid oxygen plus kerosene	SOLID → stored propellant and limited throttle control

SYSTEM / DEVICE / INSTITUTION
- SOLID → stored propellant and limited throttle control
- EARTH-STORABLE LIQUID → ambient-temperature liquid pair

DECISIVE TECHNICAL DISTINCTION
- CRYOGENIC → liquid oxygen plus liquid hydrogen
- SEMI-CRYOGENIC → liquid oxygen plus kerosene

STATUS / UNIT / EVIDENCE LIMIT
- RULE → propellant class does not prove flight status
- SOLID → stored propellant and limited throttle control

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → propellant class does not prove flight status.

09

STAGE 09

HUMAN-RATING GATE

HUMAN-RATING GATE · RELIABLE LAUNCHER · NECESSARY BUT INSUFFICIENT BASE · MARGINS AND REDUNDANCY · TOLERATE SELECTED FAILURES · HEALTH MONITORING · DETECT UNSAFE CONDITIONS · ABORT CAPABILITY

RELIABLE LAUNCHER → necessary but insufficient base → MARGINS AND REDUNDANCY → tolerate selected failures → HEALTH MONITORING → detect unsafe conditions → ABORT CAPABILITY → protect crew across ascent phases → CERTIFICATION → separate human-rated configuration

STARTING CONDITION / INPUT
- RELIABLE LAUNCHER → necessary but insufficient base
- MARGINS AND REDUNDANCY → tolerate selected failures

SCIENTIFIC OR ENGINEERING MECHANISM
- HEALTH MONITORING → detect unsafe conditions
- ABORT CAPABILITY → protect crew across ascent phases

OUTPUT / FAILURE BOUNDARY
- CERTIFICATION → separate human-rated configuration
- RELIABLE LAUNCHER → necessary but insufficient base

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ABORT CAPABILITY → protect crew across ascent phases.

10

STAGE 10

ACCESS-TO-SPACE SYSTEM

ACCESS-TO-SPACE SYSTEM · PROPULSION, STRUCTURE, AVIONICS AND GUIDANCE · REPEATABLE STAGE AND SUBSYSTEM QUALITY · INTEGRATION, SAFETY AND LAUNCH WINDOW · MISSION ASSURANCE · VERIFICATION AND ANOMALY CLOSURE · CADENCE AND CUSTOMER CONFIDENCE

VEHICLE → propulsion, structure, avionics and guidance

FACTORY → repeatable stage and subsystem quality → PAD / RANGE → integration, safety and launch window → MISSION ASSURANCE → verification and anomaly closure

CONCEPT / ARCHITECTURE
- VEHICLE → propulsion, structure, avionics and guidance
- FACTORY → repeatable stage and subsystem quality

MECHANISM / APPLICATION
- PAD / RANGE → integration, safety and launch window
- MISSION ASSURANCE → verification and anomaly closure

READINESS / STATUS LIMIT
- OUTPUT → cadence and customer confidence
- VEHICLE → propulsion, structure, avionics and guidance

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PAD / RANGE → integration, safety and launch window.

11

STAGE 11

FUTURE-CLAIM FIREWALL

FUTURE-CLAIM FIREWALL · INSTITUTION, VEHICLE, SPACECRAFT, ORBIT AND SERVICE · RETIRED, OPERATIONAL, APPROVED OR UNDER CONSTRUCTION · PROPULSION, INFRASTRUCTURE, INTEGRATION AND ASSURANCE · TECHNOLOGY TRANSFER FROM AUTHORISATION AND SAFETY · DATED OFFICIAL CAPACITY, SCHEDULE AND OUTCOME SOURCE

DEFINE → institution, vehicle, spacecraft, orbit and service → CLASSIFY → retired, operational, approved or under construction → TRACE → propulsion, infrastructure, integration and assurance → SEPARATE → technology transfer from authorisation and safety → VERIFY → dated official capacity, schedule and outcome source

SYSTEM / DEVICE / INSTITUTION
- DEFINE → institution, vehicle, spacecraft, orbit and service
- CLASSIFY → retired, operational, approved or under construction

DECISIVE TECHNICAL DISTINCTION
- TRACE → propulsion, infrastructure, integration and assurance
- SEPARATE → technology transfer from authorisation and safety

STATUS / UNIT / EVIDENCE LIMIT
- VERIFY → dated official capacity, schedule and outcome source
- DEFINE → institution, vehicle, spacecraft, orbit and service

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated official capacity, schedule and outcome source.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · STRATEGIC R&D ANCHOR AND TECHNOLOGY OWNER FOR LAUNCH-VEHICLE FAMILIES · SINGLE-WINDOW NODAL MECHANISM FOR PROMOTING, AUTHORISING AND SUPERVISING SPACE · COMMERCIAL PSU HANDLING LAUNCH SERVICES, SATELLITE SERVICES AND INDUSTRY-LINKED · SOVEREIGN POLICY UMBRELLA THAT STILL KEEPS SPACE AS A · THE REFORM QUESTION IS NOT "STATE VERSUS PRIVATE" BUT · REMAIN SOVEREIGN, WHICH CAN BE COMMERCIALLY SCALED, AND HOW · LAUNCH AUTHORISATION INVOLVES LIABILITY, RANGE SAFETY, EXPORT-CONTROL SENSITIVITY, SPECTRUM/ORBITAL

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH
- ISRO: strategic R&D anchor and technology owner for launch-vehicle families and major missions.
- IN-SPACe: single-window nodal mechanism for promoting, authorising and supervising space activities of non-government entities.
- NSIL: commercial PSU handling launch services, satellite services and industry-linked production/commercial arrangements.
- DoS: sovereign policy umbrella that still keeps space as a strategic national capability domain.

READINESS, UNIT OR STATUS LIMIT
- CAUTION: The reform question is not "state versus private" but "which function must
- remain sovereign, which can be commercially scaled, and how risk is supervised."
- CAUTION: Launch authorisation involves liability, range safety, export-control sensitivity, spectrum/orbital coordination and national-security oversight.
- it cannot be treated like ordinary industrial licensing.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
- CAUTION: Over-centralisation can slow innovation, but under-regulation can create safety and strategic vulnerabilities in a dual-use domain.
- CAUTION: The real success metric is not merely new entrants but whether the
- ecosystem can scale manufacturing, quality assurance and globally credible launch service delivery.
- CAUTION: The unfinished legal question: India regulates private space activity through executive policy (Indian Space Policy 2023) and

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels